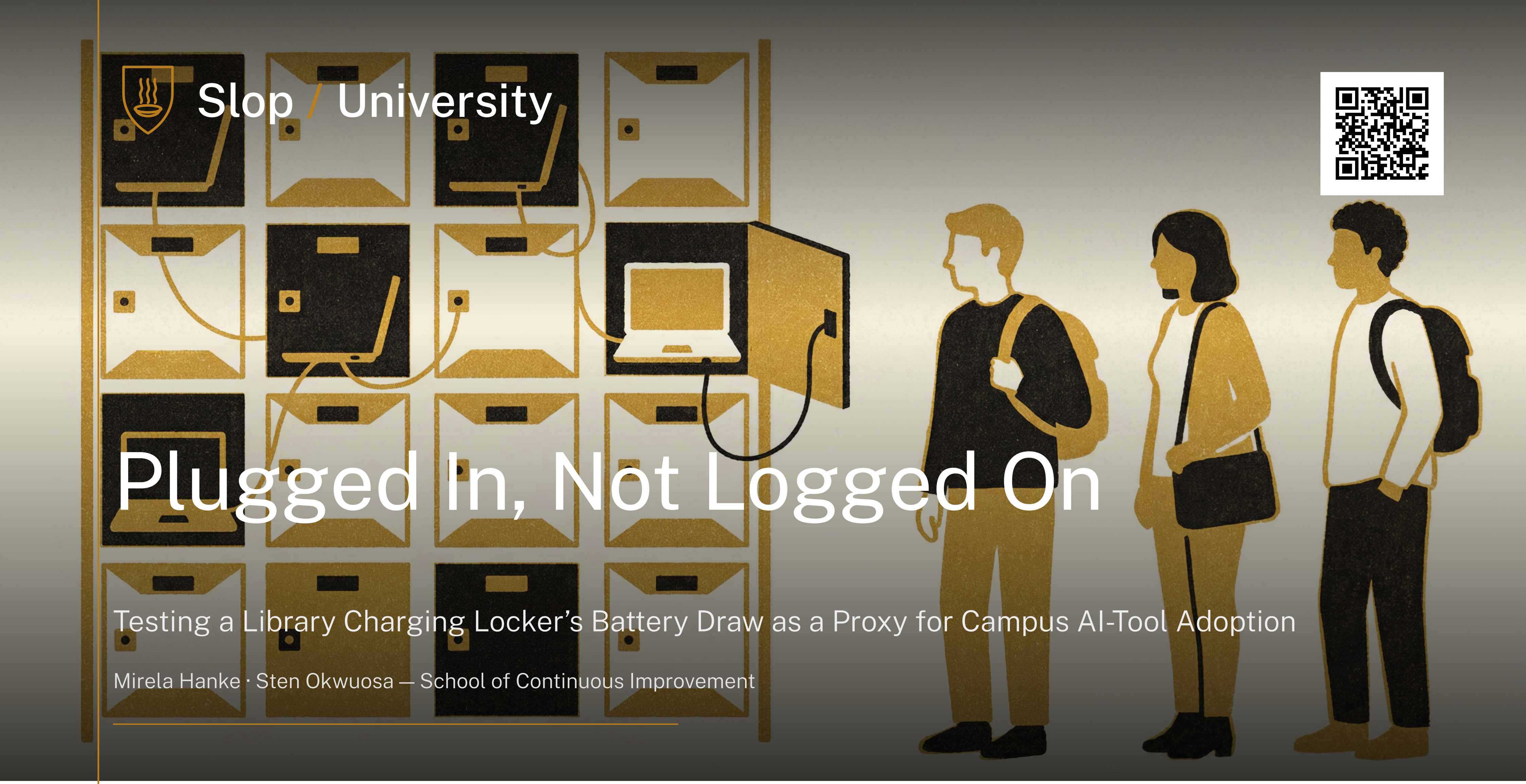

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Plugged In, Not Logged On

Testing a Library Charging Locker's Battery Draw as a Proxy for Campus AI-Tool Adoption

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Background

Library charging-locker queues have lengthened every term since agentic AI assistants moved onto students' own laptops. Sustainability now reads the lockers' aggregate battery draw as an early signal of AI-tool adoption — without asking why students are actually waiting there.

Research questions

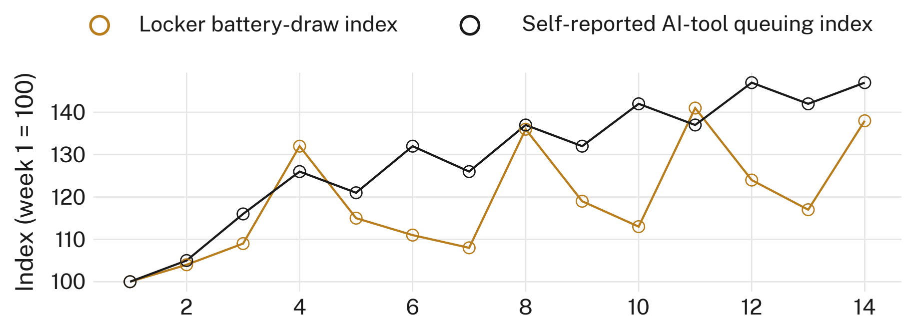
- What do students queuing at library charging lockers say they're there for?
- Does the lockers' battery-draw telemetry match those stated reasons?
- Does draw growth track AI-tool adoption, or the assessment calendar?

Approach

- n = 238 · intercept survey at 5 locker banks · 14-week term
- Locker firmware battery-draw logs, 3-min sampling, same window
- Assessment-calendar dates and AI-copilot install rate pre-registered as covariates

Findings

Self-reported AI-tool use explains little of the term's rise in locker battery draw once assessment proximity is included. Draw instead tracks the six-course assessment calendar closely throughout.



Locker battery-draw index rose 38% across the term (weeks 1–14, 2026), moving within a day of every one of six assessment deadlines (r = 0.71); self-reported AI-tool queuing explained a further 9% of variance once deadline proximity was held constant (r = 0.22).

Discussion & conclusions

A locker's charge cycle tracks a due date more reliably than whatever is running on the laptop inside it. Reading battery draw as an adoption signal risks certifying assessment stress as innovation uptake, and the indicator's clean upward trend is doing most of the persuading. Neither series was validated against actual device load before either reached a committee paper. Next: pair draw logs with device-level telemetry, and retire the index if the pairing doesn't hold, before either figure reaches a sustainability report.

Selected references

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Locker telemetry aggregated per bank; no device identifiers retained. Intercept surveys voluntary and anonymous; thresholds pre-registered.